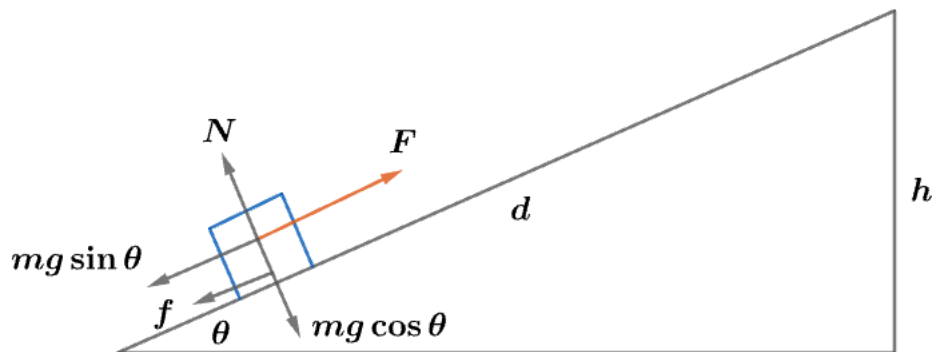


2023 F=ma Exam: Problem 4

Kevin S. Huang



The force F the person applies has to balance the component of gravity parallel to the inclined plane and kinetic friction,

$$F = mg \sin \theta + \mu_k N = mg(\sin \theta + \mu_k \cos \theta)$$

The distance d the box is dragged up is

$$d = \frac{h}{\sin \theta}$$

The work done is then given by

$$W = Fd = mgh(1 + \mu_k \cot \theta)$$

so the answer is E.